Manting Guo

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Guangdong



Education

2024 – Present **Graduate Program in Applied Mathematics, Shanghai Jiao Tong University** *(Joint program with SLAI since September 2025).*
GPA: 3.6/4.0

2020 – 2024 **B.Sc. Applied mathematics (strengthening basic disciplines program),Shanghai Jiao Tong University** *(graduated with honors).*
Thesis title: *Mathematical modeling and numerical simulation of DNA and Protein aggregation phenomena.*
In-major GPA:3.62/4.30 86.7/100 GPA:3.65/4.30 87.2/100

2014 – 2020 **Secondary education**, Guangzhou Foreign Language School

Research Experience

2021 – 2022 **Undergraduate Research Program**
Topic: Arranged Forests: Enhancing Random Forests by Reducing Feature Overlap between Trees
Advisor: prof. Yuehui Zhang
From a mathematical perspective, we explored the dual of the famous Erdős-Ko-Rado theorem. From a computer science standpoint, we proposed an algorithm leveraging the properties of prime numbers, which demonstrates greater efficiency than random forests in certain cases. We investigated k-subset selection problems within m-element matrices, where $m = p^2 + p + 1$ and p is prime. This was extended to $p \times p$ matrices under the constraint $k+l = p$, and further generalized to d-dimensional vector-valued matrices.

2023-2024 **Mathematical Modeling & Numerical Analysis for Reaction-Diffusion Systems**
Advisor: prof. Quanxing Liu
Our theoretical framework, which each response function is extracted from well-designed *in vitro* assays, offers a more general avenue to understanding phase-separation systems.
The paper 'Biomolecular condensates bridge experiment and theory of mass-conserving reaction-diffusion systems in phase separation'. *Structure, 2025, (33)* (solely responsible for the theoretical derivation and the implementation of numerical simulations and analysis.) (https://www.cell.com/structure/fulltext/S0969-2126(25)00216-3)

2024 **Research Project: Exploring scaling laws in cross-scale systems**
Advisor: Prof. Quanxing Liu
Derived scaling laws and performed bifurcation analysis in **Fortran** with **AUTO**, analyzing system stability, dynamic transitions and emergent patterns in cross-scale systems.

Research Experience (continued)

2025 – Present — **Working Papers: Agent Serving Scheduling & AI for Mathematics**
Developed AgentMIP, a DAG-aware mixed-integer programming scheduler that optimizes cross-workflow stage batching and adapter switching. ((Manuscript under review at AAAI 2027. [OpenReview])

Skills and Interests

Coding — Python, Matlab, C++ & C, R, Fortran (basic), LaTeX.

Languages — English: CET4:642 CET6:598 IELTS:7.0
German : A2

Academic interests — My main research interest lies in optimization and infrastructure for AI-agent workflows. More broadly, I am drawn to mathematically grounded problems whose impact can be observed in real-world systems, from biological pattern formation and scientific computing to future applications in AI-driven decision-making, finance, education, and other complex domains. Curious about diverse subjects.

Other interests — Fitness and weight training, guitar (Fingerstyle), model building and assembly ,ball games, hiking, …

Miscellaneous Experience

Awards and Achievements

2025 — **Outstanding Academic Scholarship, School of Mathematical Sciences**
2024 — **School level outstanding graduate.**
Department Prize for Outstanding Bachelor thesis
2023 — **Third-Class Scholarship, School of Mathematical Sciences**
2022 — **Outstanding Academic Scholarship, School of Mathematical Sciences**
2021 — **Outstanding Academic Scholarship, School of Mathematical Sciences**
School level Basketball competition, bronze medalist.
2020 — **School level Basketball competition, Second place.**

Misc.

2021-2024 — **Served as a member of the Student Affairs Center's General Affairs Department for two years, with one year as the core team member.**
2022 — **Served as the group leader of the Return-to-School Presentation project at Shanghai Jiao Tong University, responsible for organizing and leading the group's activities.**
2020-2024 — **Served as Class Monitor during undergraduate studies.**